T22 고신뢰 코어와 build-up film의 조합

Youmee Choi · Doosan Electro-Materials



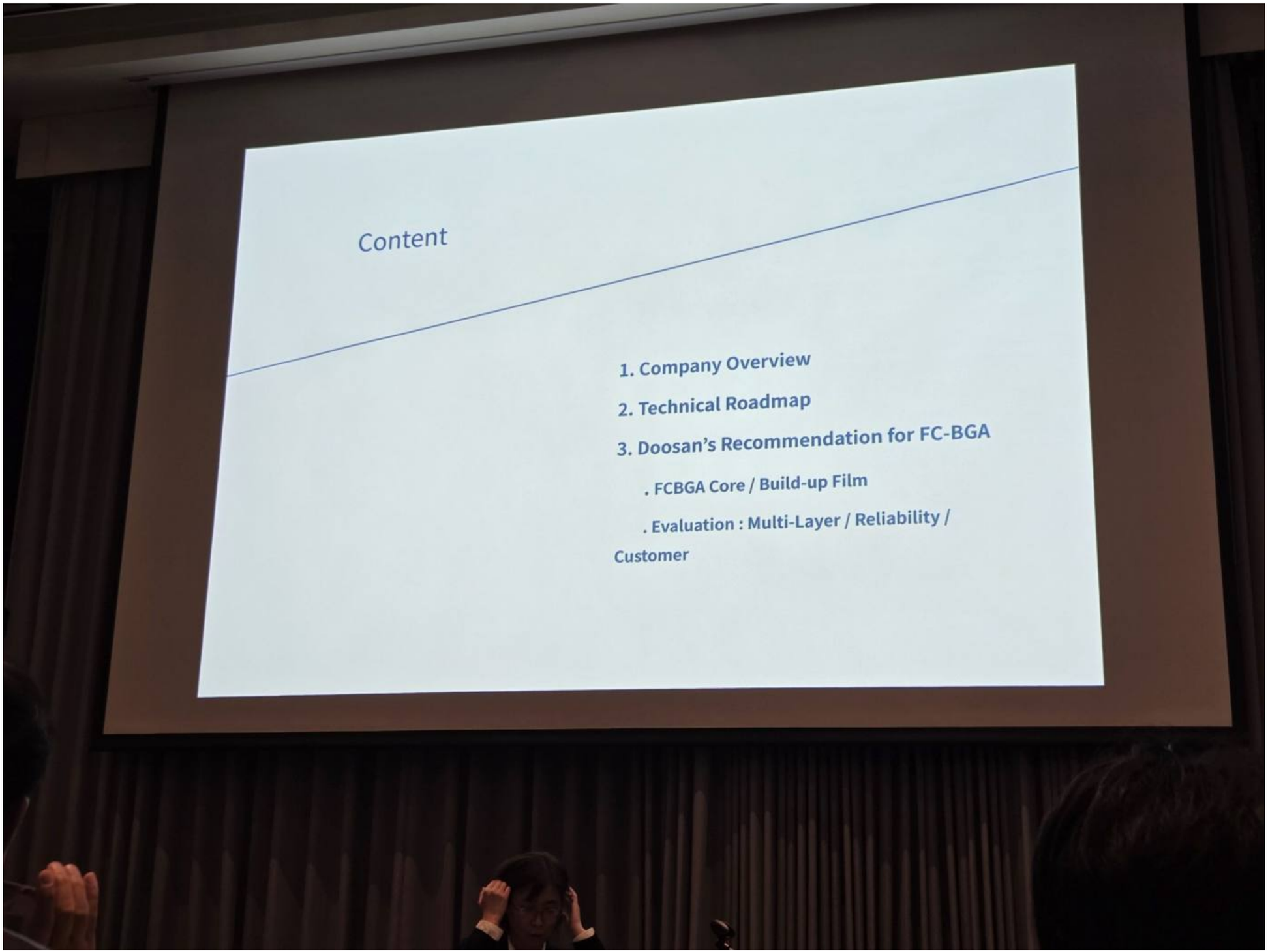
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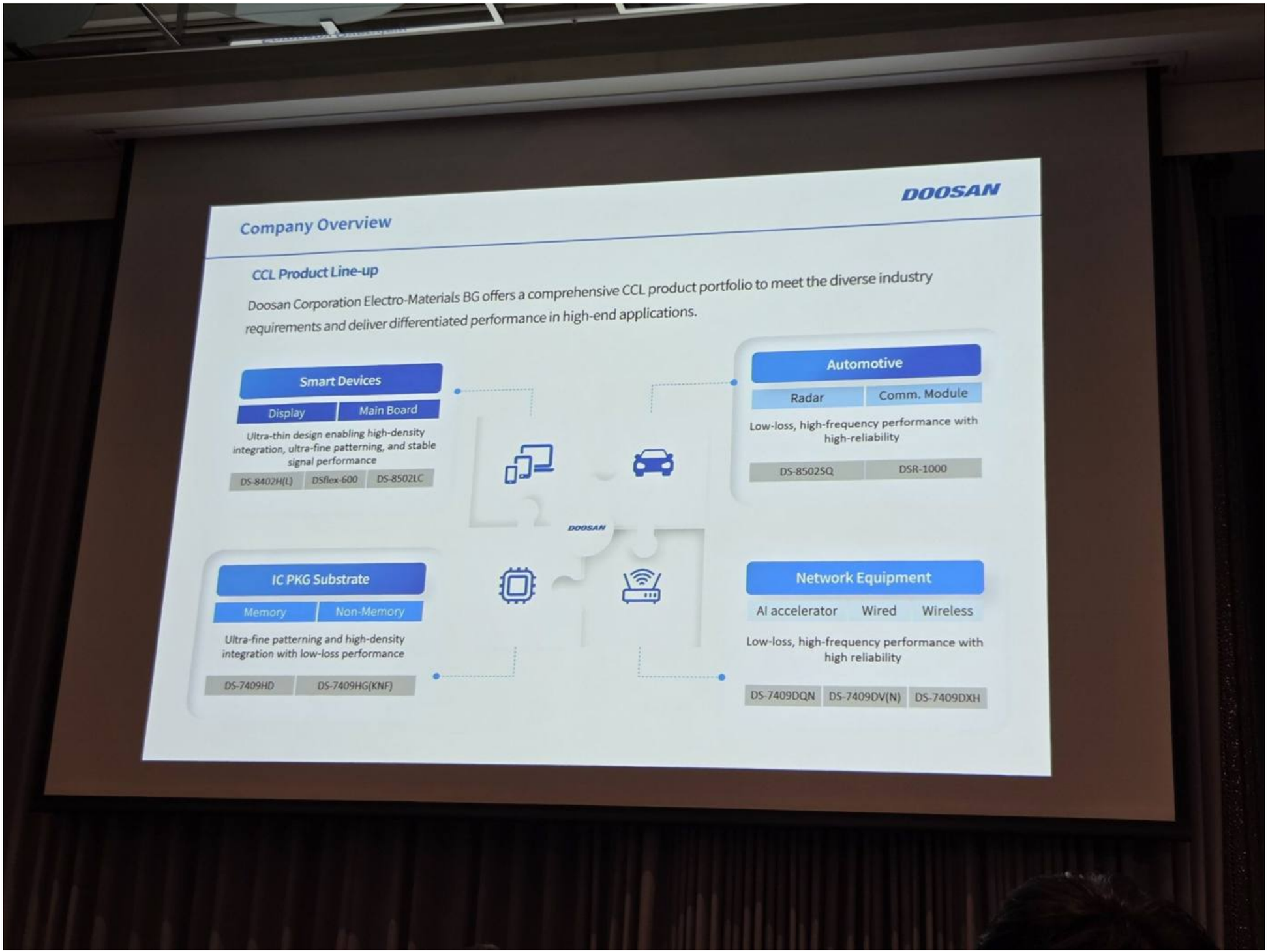
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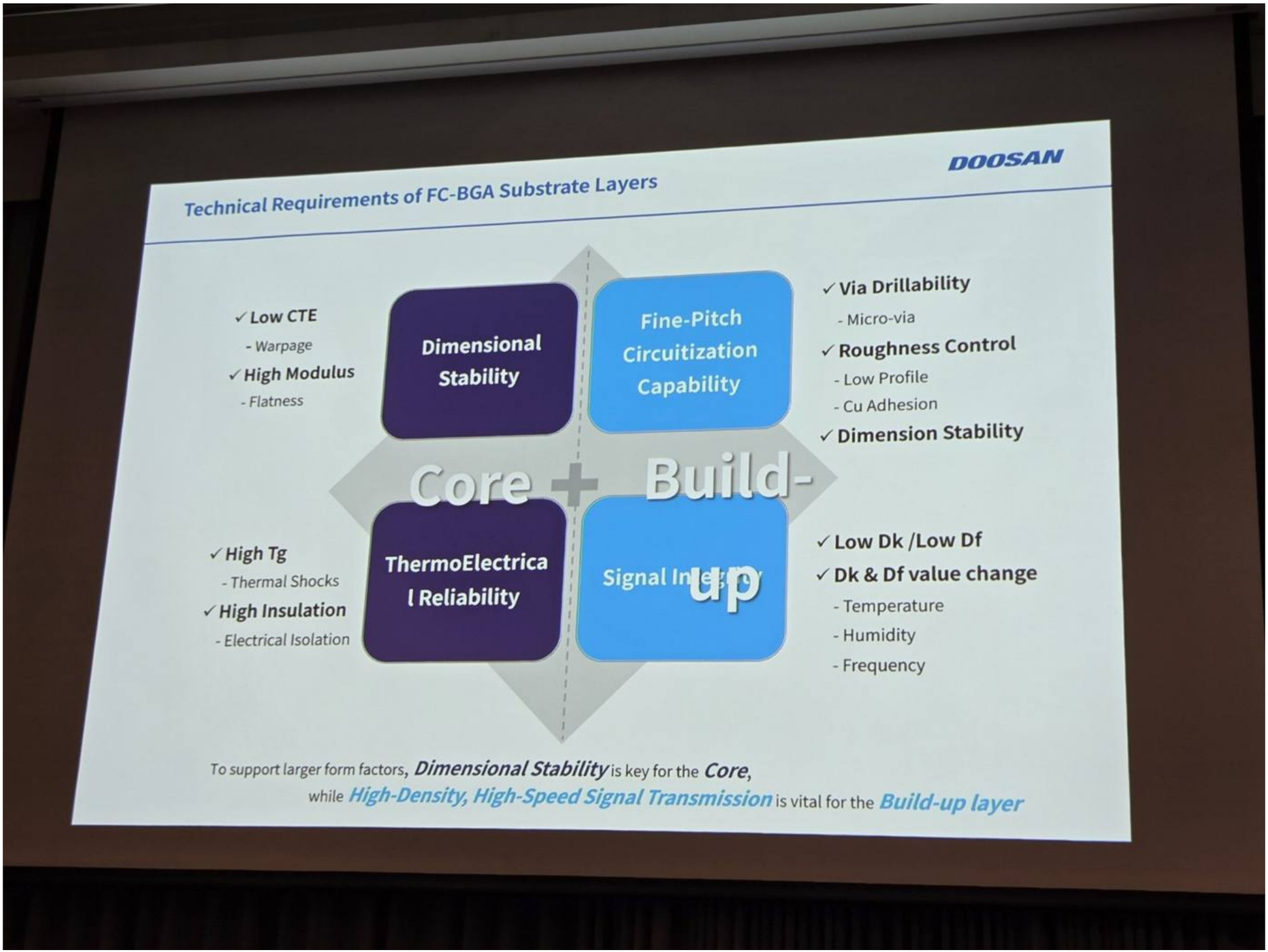
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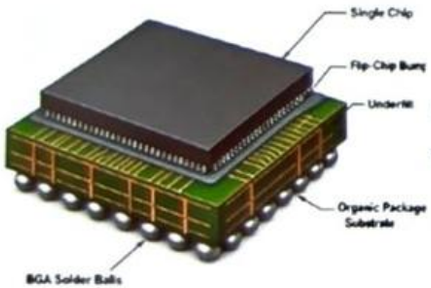
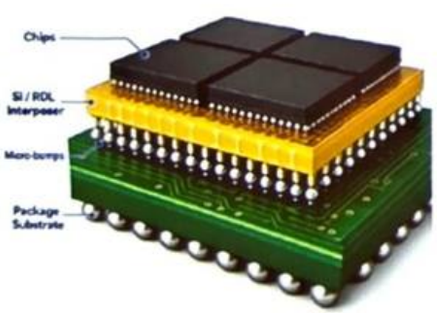
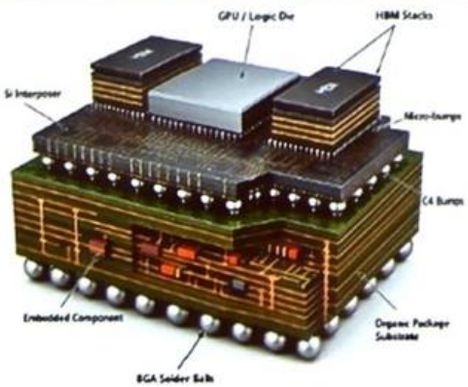
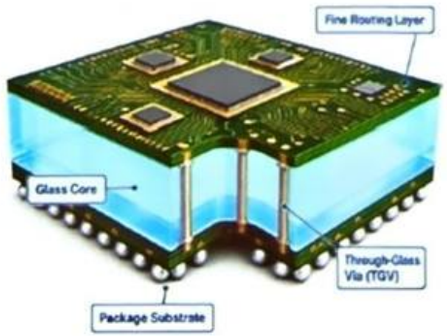
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Technical Roadmap

DOOSAN

- **Low CTE**, **High-Tg**, and **High-Modulus** Materials for Warpage Control in **Large-Area**, **High-Layer FCBGA**

Monolithic (2020)	SI/RDL Interposer (2023~25)	Embedded Component (2026~28)	Glass Core (2030~)
			
Structure - Large Single Die	- GPU + HBM + Chiplet	- Larger AI / HPC Package	- Ultra-large AI Package
Technical - Flip-Chip Bump / BGA - Organic Substrate - Yield / Warpage Control - Die Size Limit - Needs : Low Warpage	- Micro bump / TSV / C4 - Interposer + Organic Sub. - High Bandwidth / Fine Routing - Assemble Yield / Warpage - Needs : High Modulus, Low CTE	- Short Power & Signal Path - Embedded Passive - PI/SI Improvement - Area Reduction - Embedded Yield / Inspection - Needs : Build-up PPG, Lower Warpage, Low Dk/Df	- TGV / Fine Routing - Glass Core + Build-up Layer - Low Warpage - Dimensional Stability - TGV Yield / Glass Brittleness - Needs : Warpage, Low Dk/Df

T22 고신뢰 코어와 build-up film의 조합

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Recommendation for FC-BGA

DOOSAN

Total Solution

. Warpage control
. SAP processability
. High Layer Lamination

. Low CTE
. Surface Roughness
. Cu Adhesion

Core CCL (IULH)



Cu 1.5 ~ 35μm

Dielectric Layer 0.2 ~ 1.6T

Cu 1.5 ~ 35μm

Properties	Condition	Unit	IU
Glass type			Low CTE glass
Tg	DMA	°C	310
CTE	TMA	ppm/°C	5
Storage Modulus (50°C)	DMA	GPa	32
Storage Modulus (260°C)	DMA	GPa	22
Peel strength	12um VLP	Kg/cm	0.7
Dielectric Constant (Dk)	SPDR	10GHz	4.3
Dissipation Factor (Df)	SPDR	10GHz	0.010

Build-up Film (DSBF)



Protective Film

Dielectric Layer 10 ~ 50 μm

Base Film

Properties		Condition	Unit	DSBF-300
Peel Strength*		SAP Plating	kgf/cm	0.41(D25min)
Surface Roughness	Ra	After Desmear	nm	50 ~ 70
	Rz		um	0.45 ~ 0.70
CTE_x, y	α1	TMA	ppm/°C	15
Dk**@10GHz		SPDR	-	3.4
Df** @10GHz			-	0.0030~0.0035
Tg		DMA	°C	160
Elongation		UTM	%	2.2

*Applied Adhesion Promoter

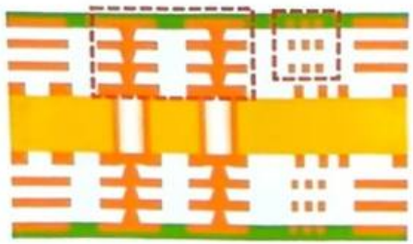
**Dk/Df Method : Split Post Dielectric Resonators[SPDR]

Evaluation : Multi-layer _ Via & Fine Pattering

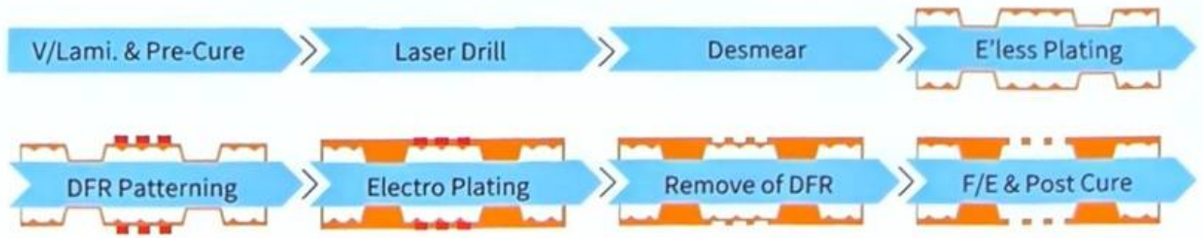
DOOSAN

. 3-2-3 Layer, **Stack Via** & **Fine Patterning** Test, 100*100㎜ unit body size

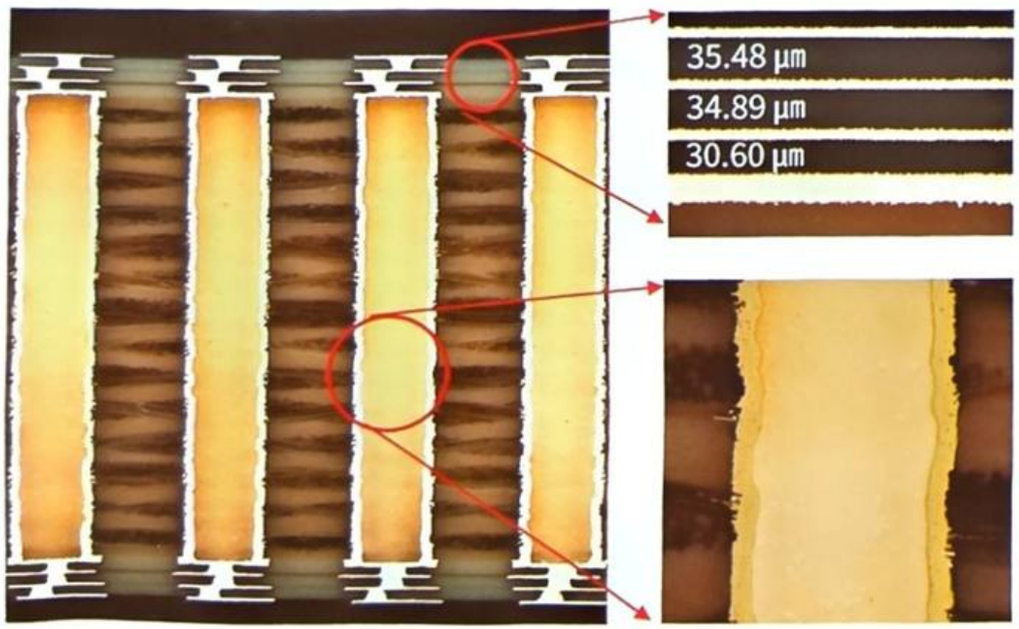
▶ Test Structure



▶ SAP Process



Test Result : Stack Via



▶ Adhesion Value with Core Layer

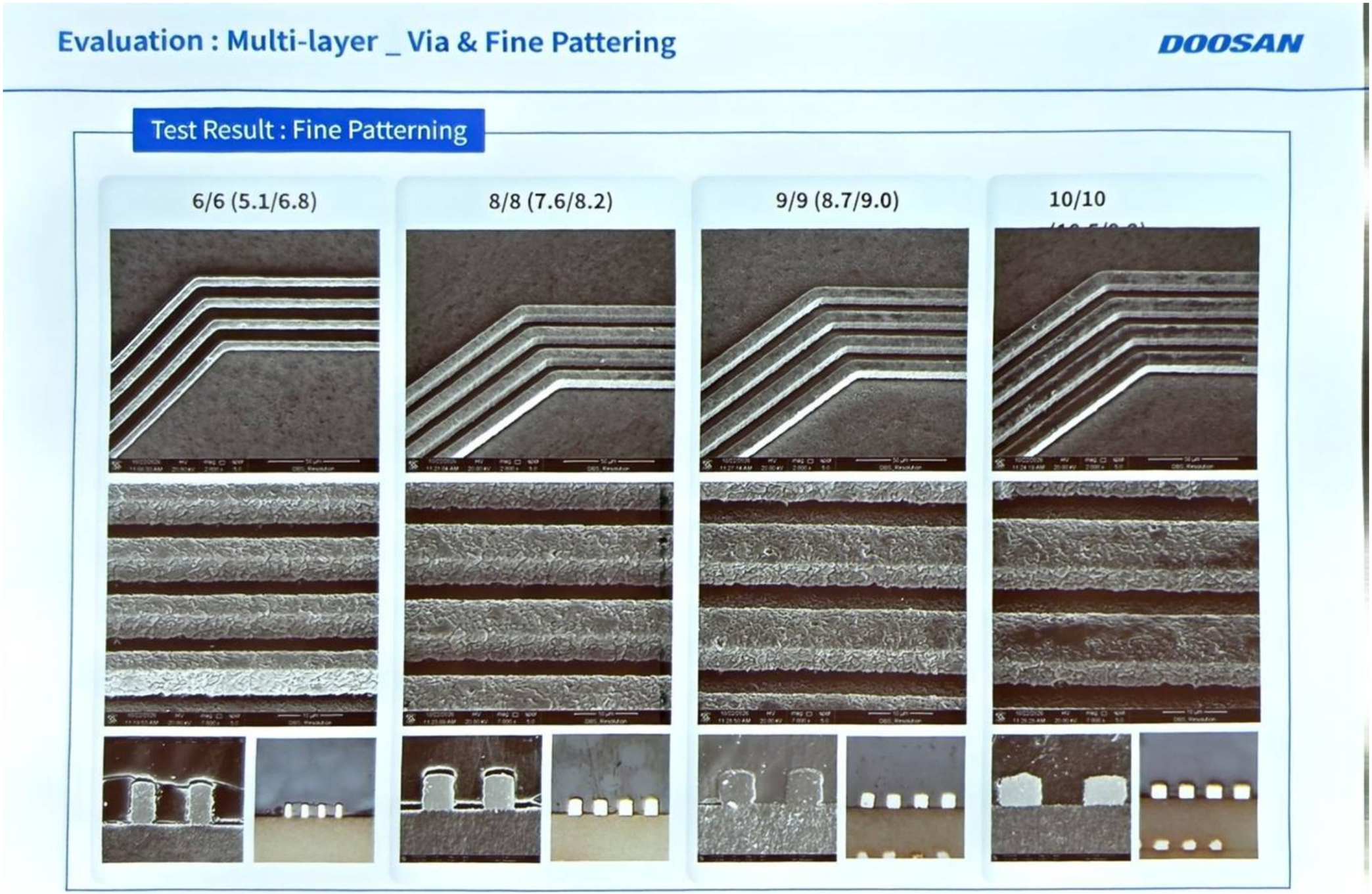
	Initial Status	B-HAST/ PCT
CZ-8101	0.62	0.61/0.62
CZ-8401	0.53	0.49/0.47

▶ Excellent THK Uniformity of DSBF-300
▶ Uniform Wall Roughness of IULH
▶ Excellent Stack Via Processability for Z-Axis Connectivity

*Via Size : 50 μm
*L/S = 6/6. 8/8, 9/9, 10/10 μm
*Cu Ratio : 75 % / 4L Cu 27um / SAP Cu 10um

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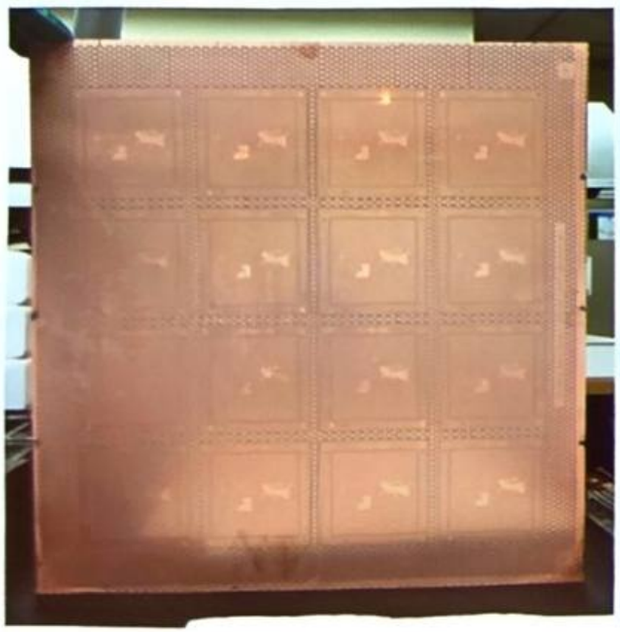


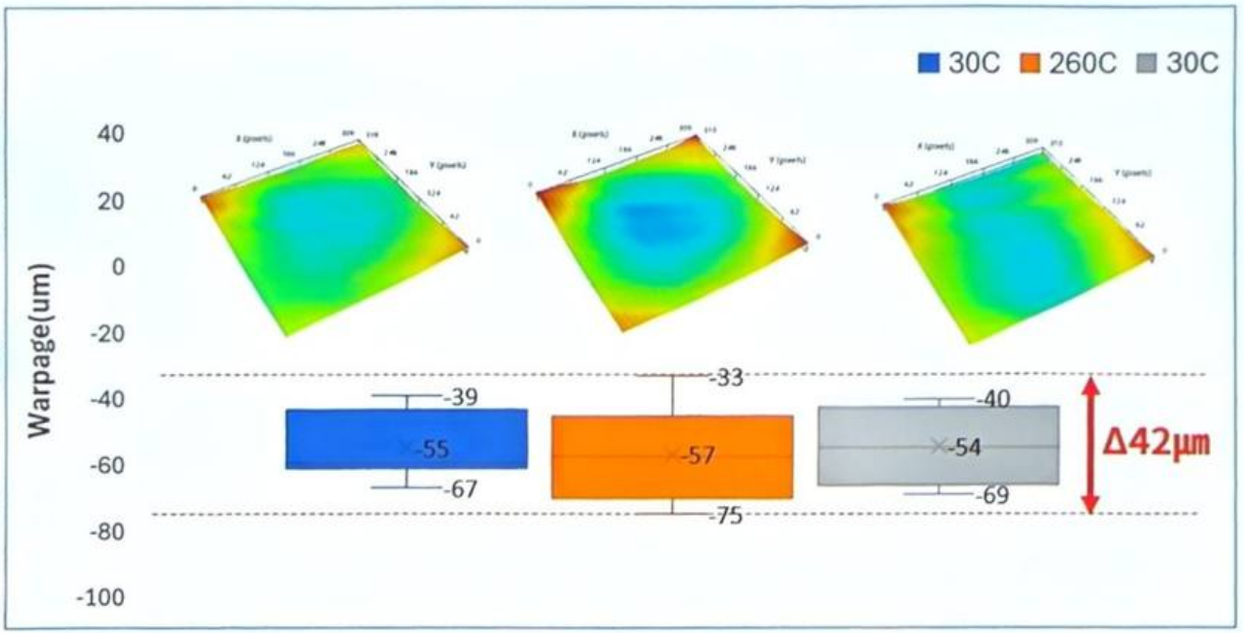
Evaluation : Multi-layer _ Warpage

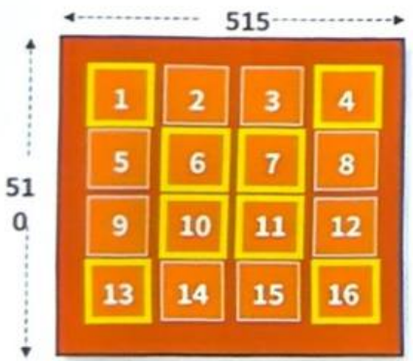
DOOSAN

- 2-2-2 Layer, **515*510 ㎜ Full Panel**, 100*100 ㎜ unit body size

▶ Test Structure







"Stable Warpage Variation with Temperature"

▶ Test Condition

: Thermal Moire

: 100*100 ㎜ Unit

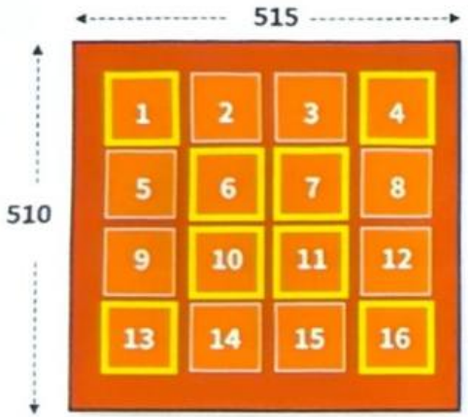
: 30°C → 260°C → 30°C

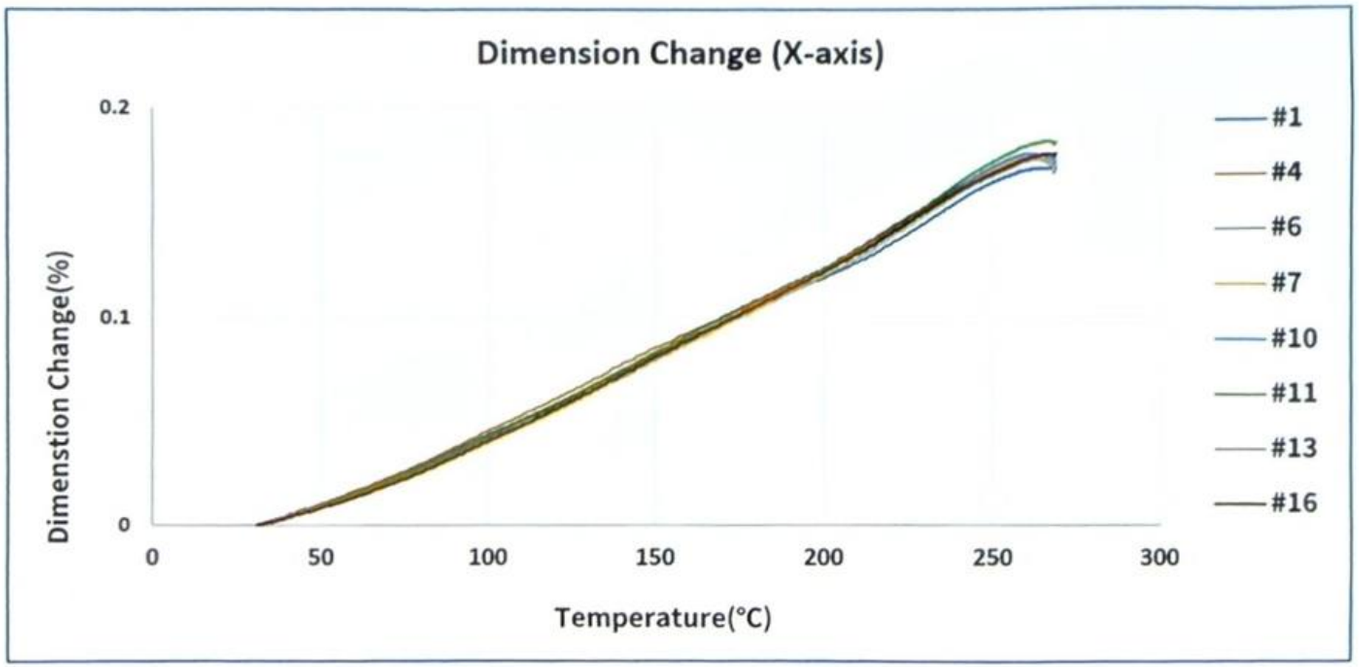
Evaluation : Multi-layer _ CTE

DOOSAN

- 2-2-2 Layer, **515*510 ㎜ Full Panel**, 100*100 ㎜ unit body size

▶ Test Structure





▶ Test Condition

: Thermal Digital Image Correlation (DIC)

: 100*100 ㎜ Unit

: 30°C → 260°C → 30°C (2cycle)

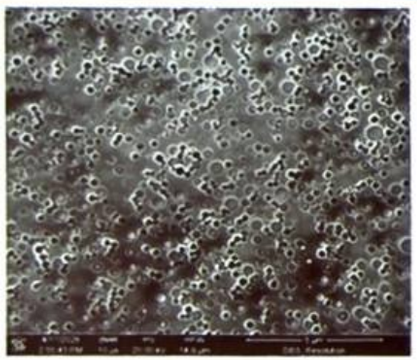
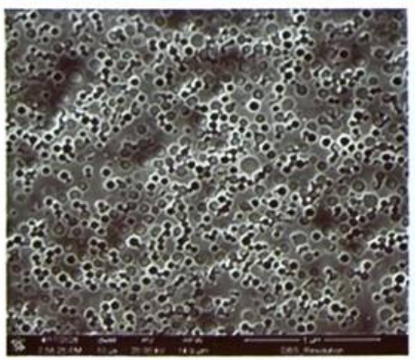
	#1	#4	#6	#7	#10	#11	#13	#16
α1	6.78	7.18	6.44	6.51	6.86	6.84	6.54	6.68
α2	8.54	8.81	9.13	9.21	9.30	9.75	9.03	8.99
Total	7.58	7.86	7.84	7.97	8.01	8.17	7.90	7.93

Evaluation : Surface Roughness of Build-up Layer

DOOSAN

- DSBF-300 Surface Roughness by Desmear Time

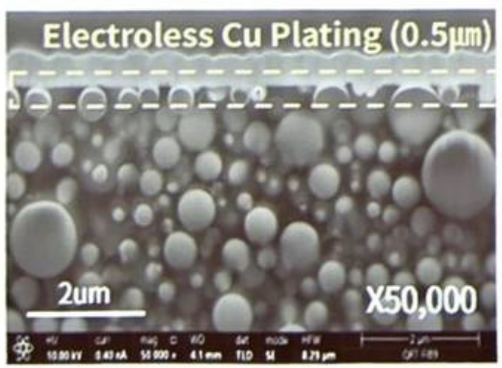
Surface Roughness

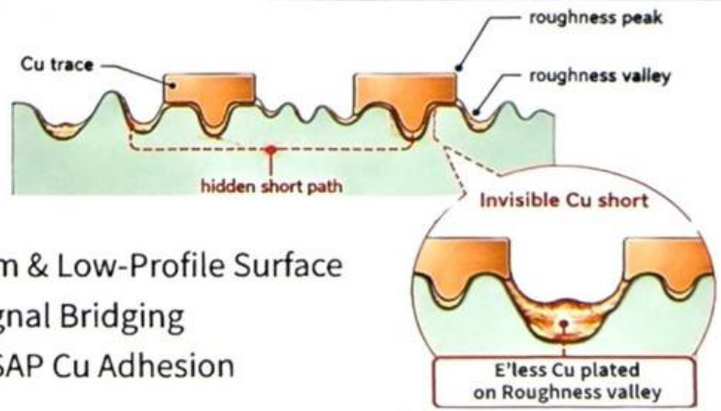
Desmear	80°C 20min	**80°C 25min**	80°C 30min
Desmear Surface Image [SEMX20K]			
Ra[nm]	20 ~ 40	**50 ~ 70**	150 ~ 200
Rz[um]	0.2 ~ 0.4	**0.45 ~ 0.7**	1.4 ~ 1.8
P/S[kgf/cm]	0.34	**0.41**	0.43

Cross-Sectional View



. Desmear Time : 25min





▶ Uniform & Low-Profile Surface
: for Signal Bridging
▶ Good SAP Cu Adhesion

Evaluation : Dk/Df Value of Build-up Layer

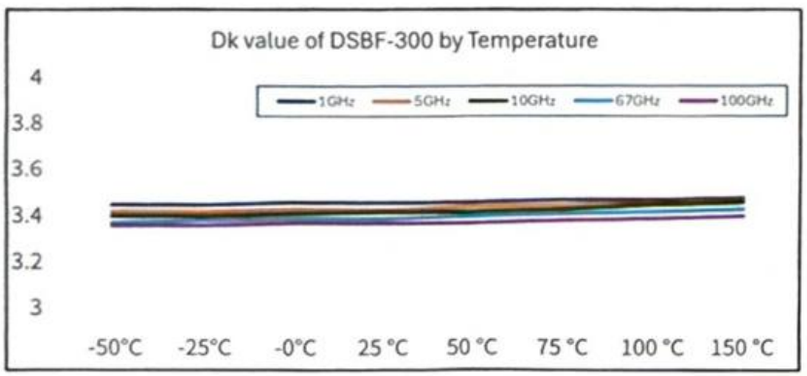
DOOSAN

- In terms of frequency change (~100GHz)

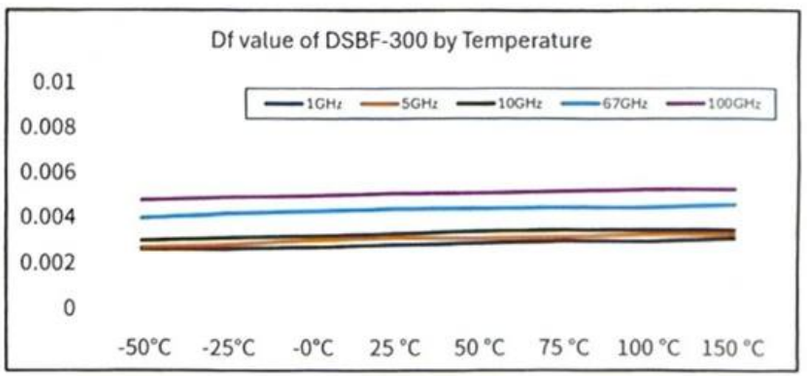
* These properties were measured with 70um [35um x 2]

	DSBF-300							
	-50°C	-25°C	0°C	25°C	50°C	75°C	100°C	150°C
1GHz	3.45 / 0.0026	3.45 0.0026	3.46 / 0.0027	3.46 / 0.0028	3.46 / 0.0029	3.47 / 0.0030	3.47 / 0.0030	3.48 / 0.0031
5GHz	3.42 / 0.0027	3.42 / 0.0028	3.43 / 0.0030	3.43 / 0.0031	3.44 / 0.0031	3.45 / 0.0032	3.46 / 0.0033	3.47 / 0.0033
10GHz	3.40 / 0.0030	3.40 / 0.0031	3.41 / 0.0032	3.42 / 0.0033	3.42 / 0.0034	3.43 / 0.0035	3.45 / 0.0035	3.46 / 0.0035
67GHz	3.37 / 0.0040	3.38 / 0.0042	3.39 / 0.0043	3.39/ 0.0044	3.40 / 0.0044	3.41 / 0.0045	3.42 / 0.0045	3.43 / 0.0046
100GHz	3.36 / 0.0048	3.36 / 0.0049	3.37 / 0.0050	3.37 / 0.0051	3.37 / 0.0051	3.38 / 0.0052	3.39 / 0.0053	3.40 / 0.0053

Dk Value



Df Value



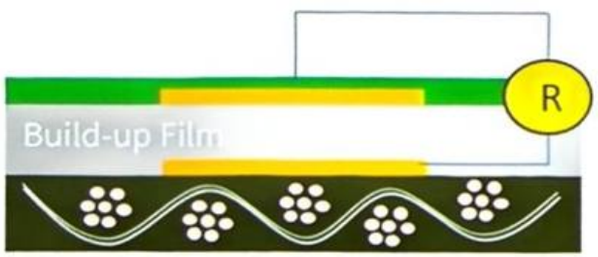
Evaluation : Reliability Test

DOOSAN

- b_HAST (Layer to Layer _ Dielectric Thickness = 25um)

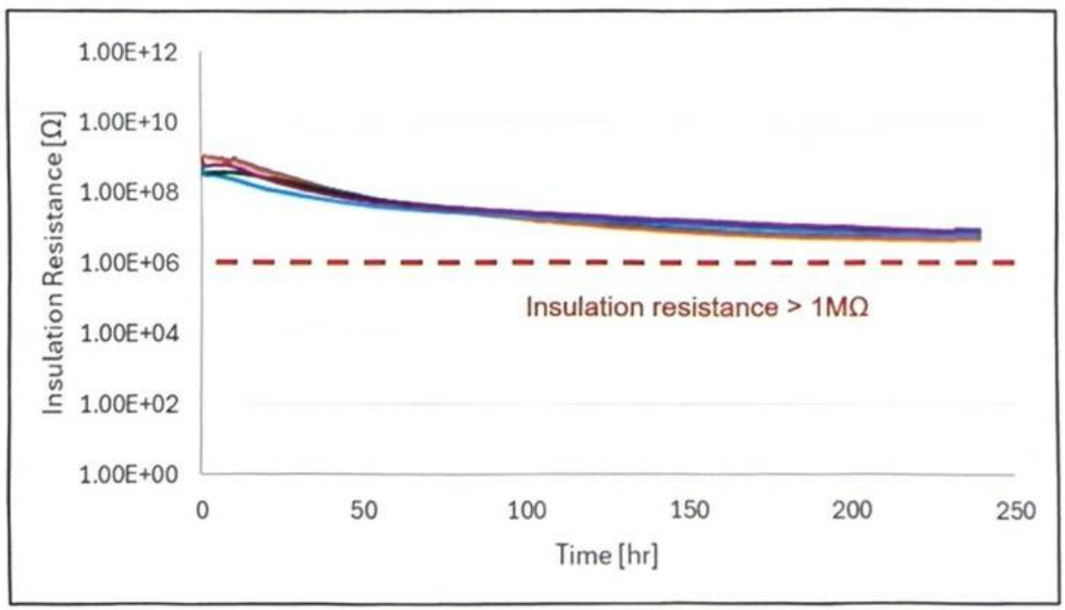
Reliability (Lab.)		Test coupon		Target Spec.	Test Result
		Coupon Design	Dielectric Thickness		DSBF-300
b_HAST	130°C, 85%, 6V, 240hrs	Layer to Layer	25 um	Non-Migration, No blister, No delamination	PASS

▶ Test Structure



- Core Th'k : 0.8T
- Build-up Film Th'k : 25um
- Cu Plating Th'k : 15 ~20um
- Pad Size : 10 x 10mm

▶ Test Results



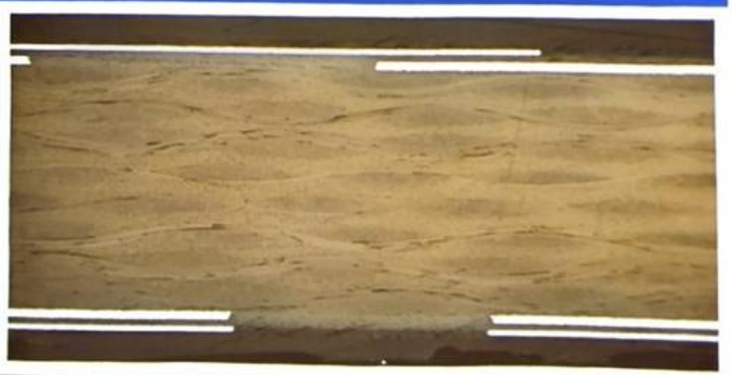
Evaluation : Reliability Test

DOOSAN

- Moisture Absorption and Heat Resistance Test

	Test	Target Spec.	Test Result
Condition	PCT 2hr → Reflow260°C(10sec), 5cycle → PCT 2hr	No blister, No delamination	PASS
	85°C, 85%, 48hr Moisture Absorption → Reflow 260°C, 20cycle		PASS

▶ Test Results

Condition	1) PCT 2hr 2) Reflow260°C(10sec), 5cycle 3) PCT 2hr	1) 85°C, 85%, 48hr 2) Reflow 260°C, 20cycle
Results: Pass [Structure] 		

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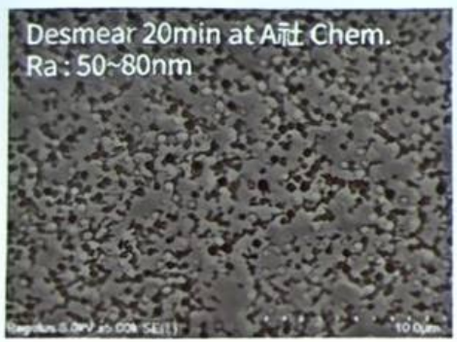
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Evaluation : Customer

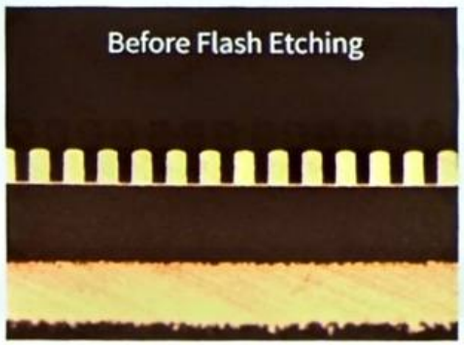
DOOSAN

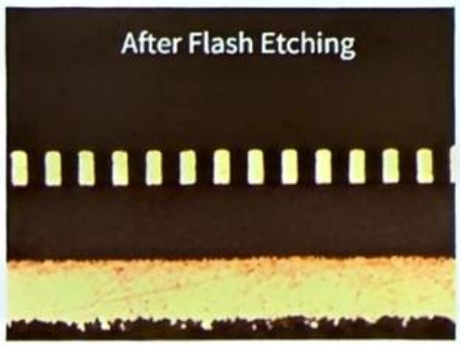
Results : Pattern

Surface



Pattern Test L/S = 7/8





- P/S = Min. 0.40kgf/cm
- Ra = 50~80nm
- Via Formation
- Smear Free

Results : Via

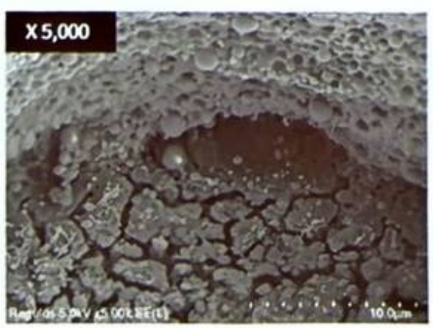
After Laser Drill



After Desmear



After Laser Drill



After Desmear



*Dry Plasma not applied

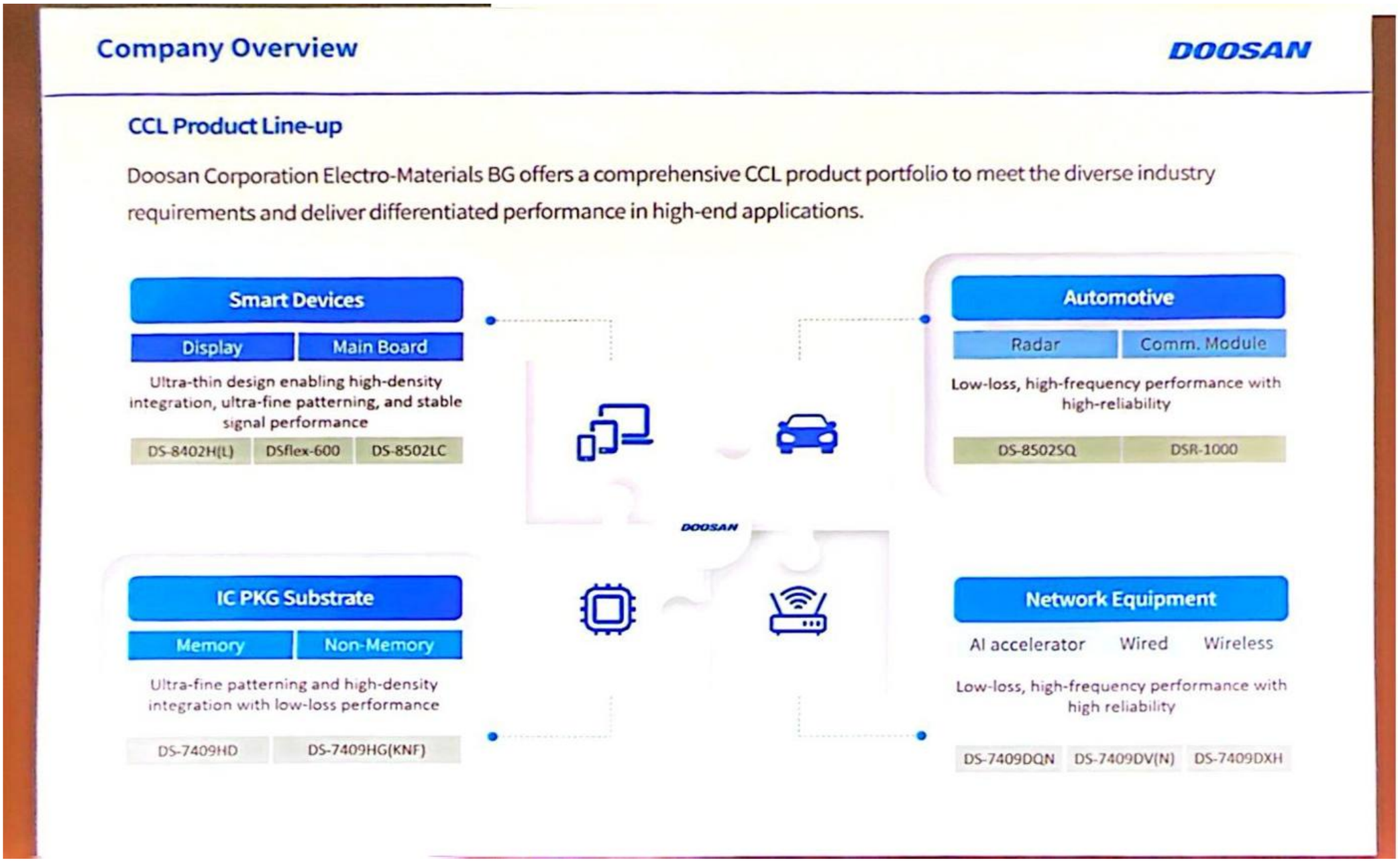
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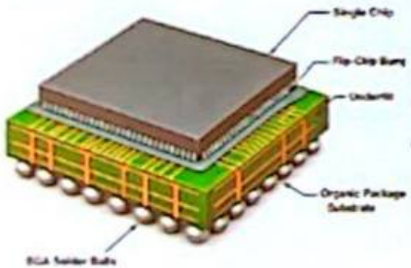
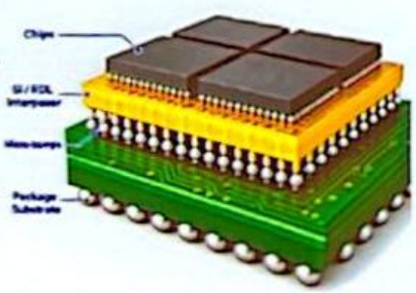
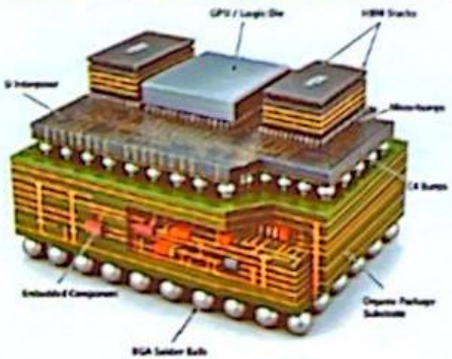
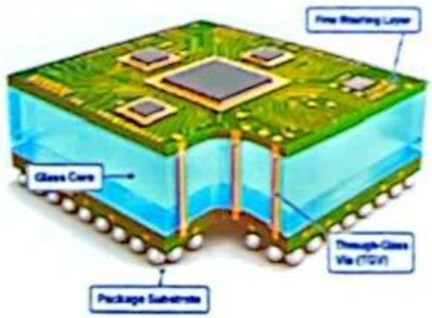
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Technical - Flip-Chip Bump / BGA - Organic Substrate - Yield / Warpage Control - Die Size Limit - Needs : Low Warpage	- Micro bump / TSV / C4 - Interposer + Organic Sub. - High Bandwidth / Fine Routing - Assemble Yield / Warpage - Needs : High Modulus, Low CTE	- Short Power & Signal Path - Embedded Passive - PI/SI Improvement - Area Reduction - Embedded Yield / Inspection - Needs : Build-up PPG, Lower Warpage, Low Dk/Df	- TGV / Fine Routing - Glass Core + Build-up Layer - Low Warpage - Dimensional Stability - TGV Yield / Glass Brittleness - Needs : Warpage, Low Dk/Df

Recommendation for FC-BGA

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Total Solution

. Warpage control
. SAP processability
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Dielectric Layer 0.2 ~ 1.6T

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Properties	Condition	Unit	IU
Glass type			Low CTE glass
Tg	DMA	°C	310
CTE	TMA	ppm/°C	5
Storage Modulus (50°C)	DMA	GPa	32
Storage Modulus (260°C)	DMA	GPa	22
Peel strength	12um VLP	Kg/cm	0.7
Dielectric Constant (Dk)	SPDR	10GHz	4.3
Dissipation Factor (Df)	SPDR	10GHz	0.010

Build-up Film (DSBF)



Protective Film

Dielectric Layer 10 ~ 50 μm

Base Film

Properties		Condition	Unit	DSBF-300
Peel Strength*		SAP Plating	kgf/cm	0.41(D25min)
Surface Roughness	Ra	After Desmear	nm	50 ~ 70
	Rz		um	0.45 ~ 0.70
CTE_x, y	α1	TMA	ppm/°C	15
Dk**@10GHz		SPDR	-	3.4
Df** @10GHz			-	0.0030~0.0035
Tg		DMA	°C	160
Elongation		UTM	%	2.2

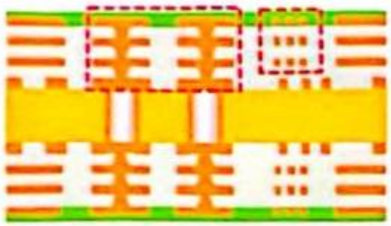
*Applied Adhesion Promoter
**Dk/Df Method : Split Post Dielectric Resonators[SPDR]

Evaluation : Multi-layer _ Via & Fine Pattering

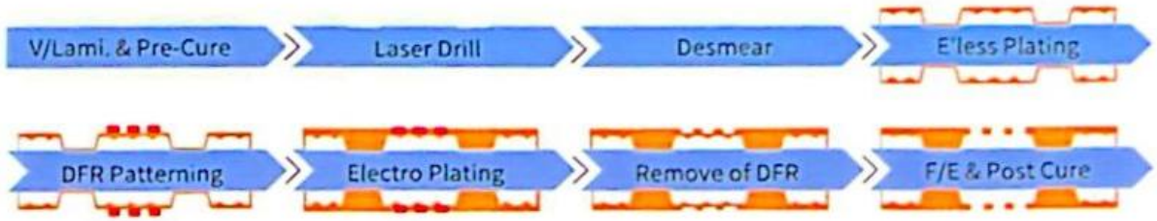
DOOSAN

. 3-2-3 Layer, **Stack Via** & **Fine Patterning** Test, 100*100㎜ unit body size

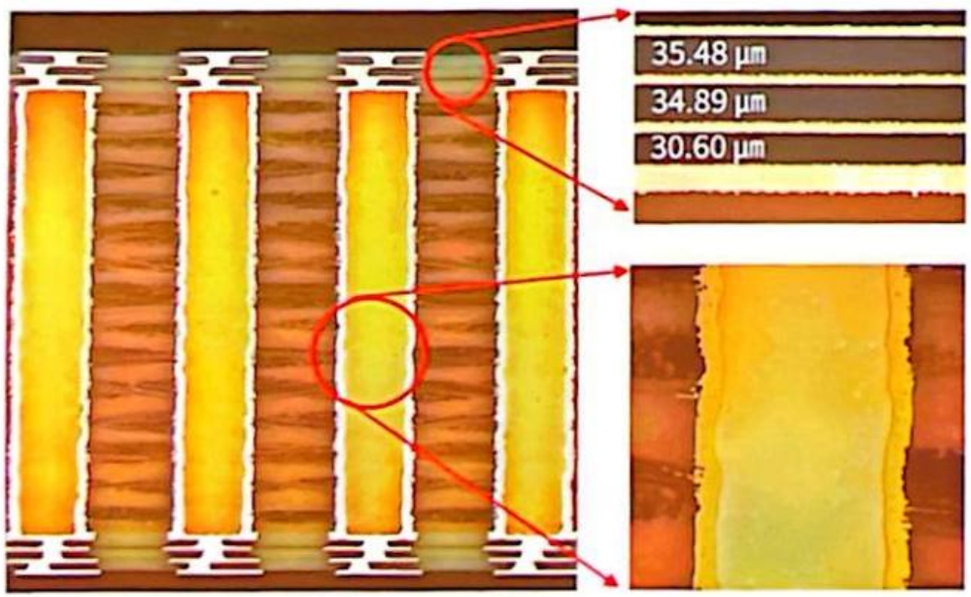
▶ Test Structure



▶ SAP Process



Test Result : Stack Via



▶ Adhesion Value with Core Layer

	Initial Status	B-HAST/ PCT
CZ-8101	0.62	0.61/0.62
CZ-8401	0.53	0.49/0.47

▶ Excellent THK Uniformity of DSBF-300
▶ Uniform Wall Roughness of IULH
▶ Excellent Stack Via Processability for Z-Axis Connectivity

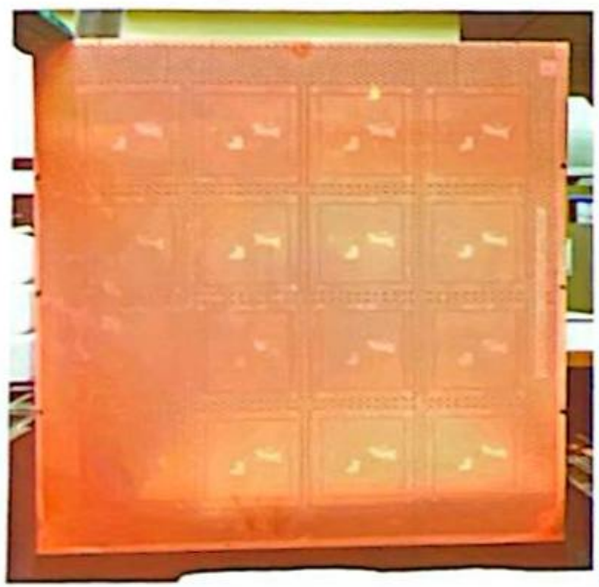
*Via Size : 50 μm
*L/S = 6/6. 8/8, 9/9, 10/10 μm
*Cu Ratio : 75 % / 4L Cu 27um / SAP Cu 10um

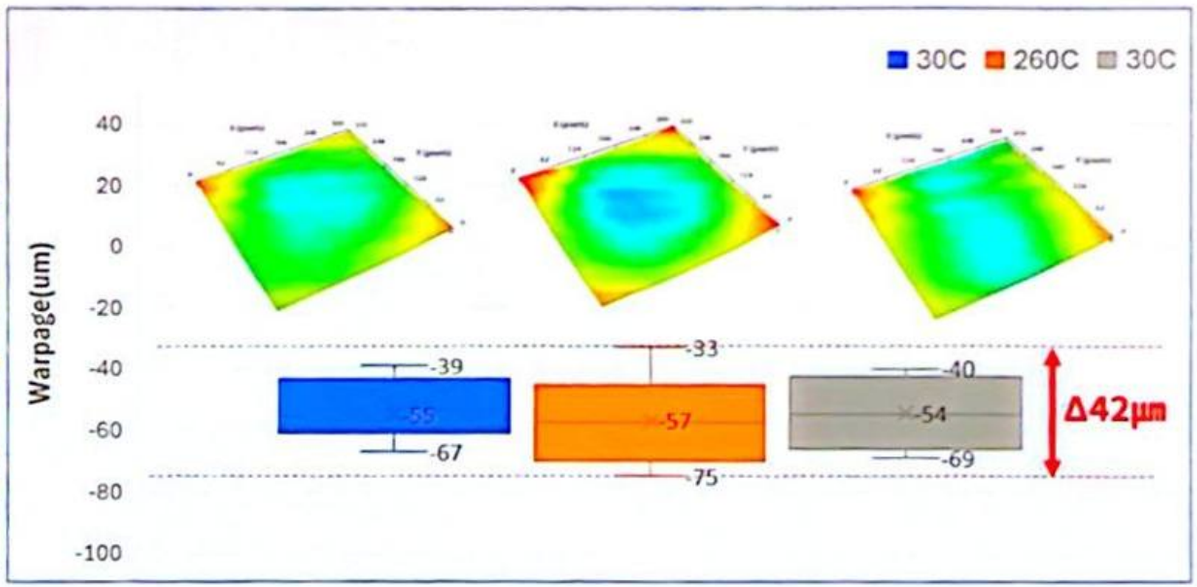
Evaluation : Multi-layer _ Warpage

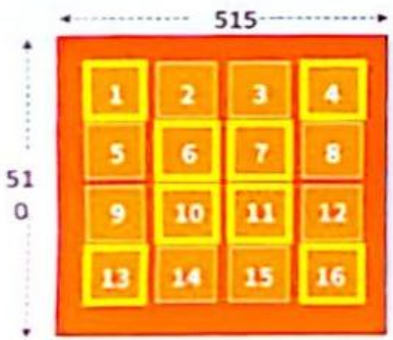
DOOSAN

- 2-2-2 Layer, **515*510 ㎜ Full Panel**, 100*100 ㎜ unit body size

▶ Test Structure







"Stable Warpage Variation with Temperature"

▶ Test Condition

: Thermal Moire

: 100*100 ㎜ Unit

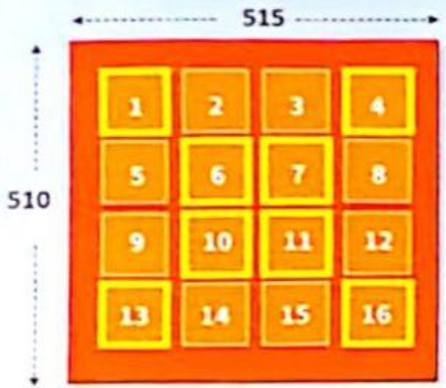
: 30°C → 260°C → 30°C

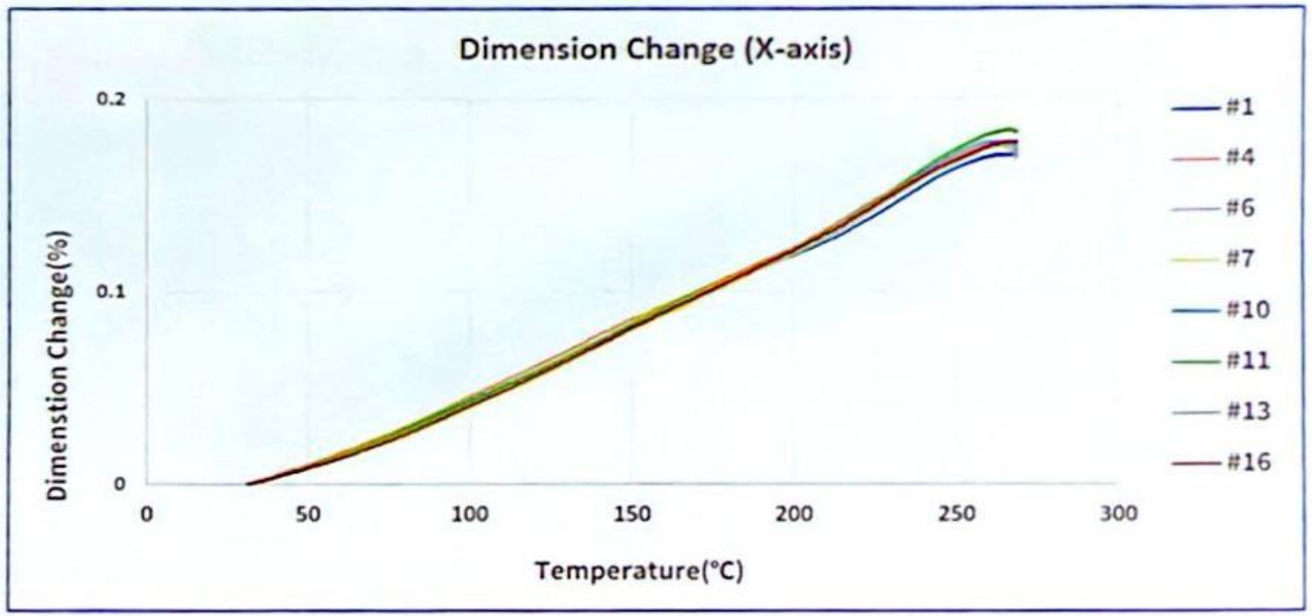
Evaluation : Multi-layer _ CTE

DOOSAN

- 2-2-2 Layer, **515*510 ㎜ Full Panel**, 100*100 ㎜ unit body size

▶ Test Structure





▶ Test Condition

: Thermal Digital Image Correlation (DIC)

: 100*100 ㎜ Unit

: 30°C → 260°C → 30°C (2cycle)

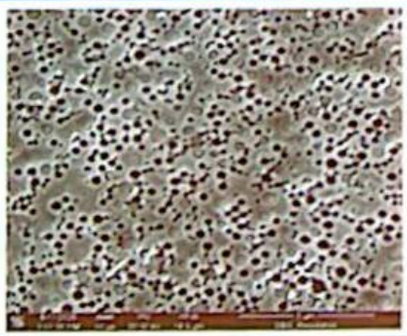
	#1	#4	#6	#7	#10	#11	#13	#16
α1	6.78	7.18	6.44	6.51	6.86	6.84	6.54	6.68
α2	8.54	8.81	9.13	9.21	9.30	9.75	9.03	8.99
Total	7.58	7.86	7.84	7.97	8.01	8.17	7.90	7.93

Evaluation : Surface Roughness of Build-up Layer

DOOSAN

- DSBF-300 Surface Roughness by Desmear Time

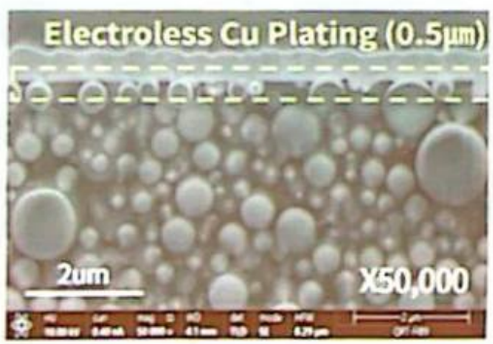
Surface Roughness

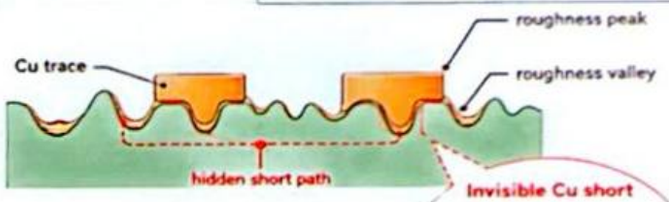
Desmear	80°C 20min	80°C 25min	80°C 30min
Desmear Surface Image [SEMX20K]			
Ra[nm]	20 ~ 40	**50 ~ 70**	150 ~ 200
Rz[um]	0.2 ~ 0.4	**0.45 ~ 0.7**	1.4 ~ 1.8
P/S[kgf/cm]	0.34	**0.41**	0.43

Cross-Sectional View

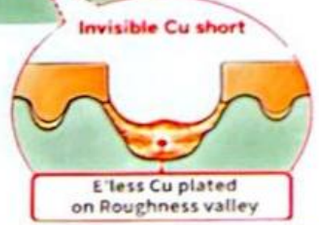
E'less Plating

. Desmear Time : 25min





▶ Uniform & Low-Profile Surface
: for Signal Bridging
▶ Good SAP Cu Adhesion



T22 고신뢰 코어와 build-up film의 조합

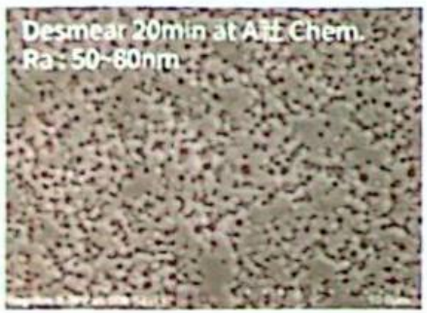
Youmee Choi · Doosan Electro-Materials

Evaluation : Customer

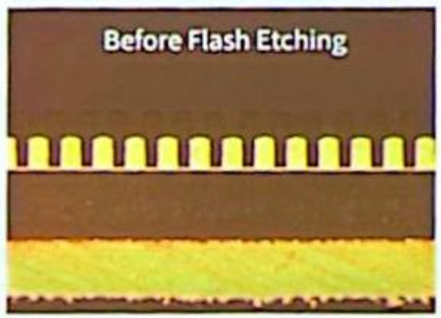
DOOSAN

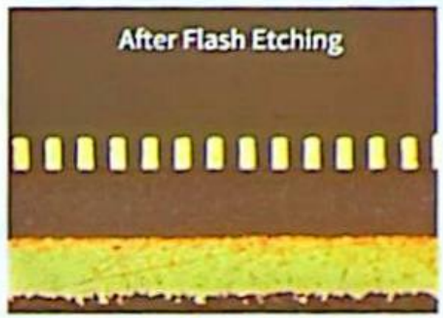
Results : Pattern

Surface



Pattern Test L/S = 7/8





- P/S = Min. 0.40kgf/cm
- Ra = 50~80nm
- Via Formation
- Smear Free

Results : Via

After Laser Drill



After Desmear



After Laser Drill



After Desmear



*Dry Plasma not applied